G.M.M MIFTAHUL ALAM ADIB

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EDUCATION

B.Sc. in Statistics Nov 2024 – Present
Shahjalal University of Science and Technology, Sylhet

SKILLS AND INTERESTS

Programming Languages	Python, SQL, C++, HTML
Machine Learning	PyTorch, TensorFlow, Scikit-Learn, Hugging Face, YOLO, LangChain, Optuna
Databases	PostgreSQL, FAISS (Vector Database)
Data Visualization	Pandas, Matplotlib, Seaborn
Web & DevOps	FastAPI, Pydantic, Streamlit, Docker, GitHub
Interests	AI Competitions, Agentic AI

COMPETITIONS

Shobdotori: Where Dialects Flow into Bangla * 2025

- In this AI competition our team **Backprop Sust** secured **4th position among 60+ teams** by developing a dual-stage fine-tuning pipeline using Whisper Medium and LoRA.

RESEARCH PUBLICATION

A Dual Pipeline Machine Learning Framework for Automated Multi-Class Sleep Disorder Screening (Under Review) Jan 2026

- Designed a dual-pipeline architecture to model physiological dependencies.
- Applied **SMOTETomek** resampling to address class imbalance and achieved **98.67% accuracy**, outperforming baseline models, validated using the Wilcoxon Signed-Rank Test.
- **Links:** GitHub * | arXiv *

A Coverage Preserving Ensemble Framework with Minority Recovery for Robust Indoor Localization 2026

- Developed a BLE-based indoor localization framework for nursing care environments using **Random Forest classification**, temporal smoothing, adaptive thresholding, and minority room recovery.
- The paper was also presented in the **ABC 2026 Conference**.
- **Links:** GitHub * | Paper *

PROJECTS

RAG System for Financial Reports * May 2024

- Engineered a **RAG pipeline** to answer natural language queries from financial reports using LangChain, FAISS, and PyMuPDF.

Data Analysis of Top 100 YouTube Channels * Dec 2024

- Conducted exploratory data analysis to visualize growth trends and content performance utilizing Pandas and Seaborn.

Retrieval-Augmented QA System * June 2025

- Developed a context-aware question-answering pipeline using a **FAISS vector database** and Google's TAPAS model.

Roadside Object Detection * — Aug 2025

- Built a **real-time object detection system** for road environment analysis using YOLOv11n, OpenCV, and PyTorch.

ORGANIZATIONS

Executive Member — Chattogram Forum, SUST — Oct 2025 – Present

General Member — SUST Data Science Club — Feb 2025 – Present

EXTRA-CURRICULAR

- **Notebook Expert, Kaggle** * — Jan 2025 – Present
  Recognized for high-quality notebooks and strong community engagement.
- **Junior Python Mentor, PyNEXT** — Jul 2025 – Aug 2025
  Mentored learners and supported skill development during a 7-day Python camp organized by the SUST Data Science Club.
- **Field Data Collection: National Parliamentary Election Survey** — Jan 2026
- **Supervised Machine Learning: Regression and Classification** *